

CIS41032: Practical for Advance Database Management System

Department of ICT

Faculty of Technology

South Eastern University of Sri Lanka

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Lab Sheet 09

Title: Work with Query and Projection Operators.

Objective:

- Specify Conditions Using Query Operators.
- Specify AND, OR, NOT Conditions.

Comparison

Equal	{<key>: {\$eq :<value>}}
Greater than	{<key>: {\$gt:<value>}} {<key>: {\$gte:<value>}}
Greater than equal	
Values in an array	{<key>: {\$in:[<value1>, <value2>,...<valueN>]}}
Less than	{<key>: {\$lt:<value>}}
Less than equal	{<key>: {\$lte:<value>}}
Not equal	{<key>: {\$ne:<value>}}

Logical

```

$and db.collectionName.find({ $and: [ {key1:value1}, { key2:value2} ] })

$or db.collectionName.find({ $or: [ {key1:value1}, { key2:value2} ] })

$not db.collectionName.find({ $not: [ {key1:value1}, { key2:value2} ] })

```

Database: CompanyData**Collection:** inventory

Item	Quantity	Status	Category	Supplier	Price	Location
Note Book	25	A	Stationery	ABC Traders	45.00	Shelf-A
Journal	50	B	Stationery	XYZ Ltd	85.00	Shelf-B
Paper	100	A	Office Use	ABC Traders	15.00	Shelf-C
Article	36	C	Educational	LearnFast	120.0	Shelf-A
Post Card	81	D	Gifting	PrintMate	25.00	Shelf-B
Pen	60	A	Stationery	ABC Traders	10.00	Shelf-C
Eraser	20	A	Stationery	XYZ Ltd	5.00	Shelf-A
Marker	75	B	Office Use	WriteIt	30.00	Shelf-D
File	90	B	Office Use	ABC Traders	55.00	Shelf-E
Ruler	10	A	Stationery	Tools & Co	8.00	Shelf-A
Envelope	45	C	Mailing	XYZ Ltd	20.00	Shelf-B
Highlighter	55	A	Stationery	LearnFast	28.00	Shelf-C

1. Select all the documents in the inventory collection where the status equals “A”.

```
db.inventory.find({ status: "A" })
```

2. Return all the documents where the quantity is less than 30.

```
db.inventory.find({ quantity: { $lt: 30 } })
```

- 3. Return all the documents where the status equals “A” and quantity is less than 30.**

```
db.inventory.find({
  $and: [
    { status: "A" },
    { quantity: { $lt: 30 } }
  ]
})
```

- 4. Select all the documents where the quantity is greater than 50 and the item is Post Card.**

```
db.inventory.find({
  $and: [
    { quantity: { $gt: 50 } },
    { item: "Post Card" }
  ]
})
```

- 5. Select the documents where the quantity is not equal to 100.**

```
db.inventory.find({ quantity: { $ne: 100 } })
```

- 6. Select the documents where either status is “A” or the quantity is not greater than 70.**

```
db.inventory.find({
  $or: [
    { status: "A" },
    { quantity: { $lte: 70 } }
  ]
})
```

Exercise**Database:** LibraryDB**Collection:** libraryBooks

BookID	Title	Author	Category	Copies	PublishedYear	Status
B001	Introduction to Databases	C.J. Date	Database	5	2018	Available
B002	Learning Python	Mark Lutz	Programming	2	2019	Borrowed
B003	MongoDB Basics	Kristina Chodorow	Database	8	2020	Available
B004	Clean Code	Robert C. Martin	Programming	4	2008	Available
B005	Web Development Handbook	Jon Duckett	Web	3	2017	Borrowed
B006	JavaScript: The Good Parts	Douglas Crockford	Web	7	2009	Available
B007	React Up & Running	Stoyan Stefanov	Web	1	2021	Borrowed
B008	Advanced Java	Herbert Schildt	Programming	6	2016	Available
B009	Cloud Computing Concepts	Rajkumar Buyya	Cloud	4	2022	Borrowed
B010	Node.js in Action	Mike Cantelon	Programming	3	2015	Available
B011	Fundamentals of AI	Stuart Russell	AI	2	2020	Available
B012	Machine Learning Basics	Tom Mitchell	AI	5	2021	Borrowed

1. Select all books in the "Programming" category.
2. Return all books that have fewer than 4 copies.
3. Select all books that are "Available" and published after 2015.
4. Find all books with the word "Basics" in the title.
5. Return books that were published before 2010 or have more than 5 copies.
6. Select books where the category is not "Web".
7. Return books where the author is either "Mark Lutz" or "Robert C. Martin".
8. Find all books with "Borrowed" status and fewer than 3 copies.
9. Return all books with PublishedYear between 2016 and 2020 (inclusive).